

Meshtastic Position Data Flow

How positions get from devices in the field to the Aervyx live tracking map.

Overview

There are **3 paths** a position can take from a Meshtastic device to the live map:

Path	Trigger	Internet needed on	Latency
1 - Phone GPS	Phone location services	Pilot's phone (cellular)	~1s
2 - BLE Relay	Mesh packets arrive via BLE	Pilot's phone (cellular)	~2-5s
3 - MQTT	Gateway publishes to broker	WiFi/Ethernet gateway node	~5-15s

```
flowchart TB
    subgraph field["In the Field"]
        direction LR

        subgraph pilotA["Pilot A -- no cell service"]
            A_DEV["Meshtastic\nDevice A"]
            A_PH["Phone A\nno internet"]
        end

        subgraph pilotB["Pilot B -- has cell service"]
            B_DEV["Meshtastic\nDevice B"]
            B_PH["Phone B\ncell + BLE"]
        end

        subgraph gw["Stationary WiFi Gateway"]
            GW_DEV["Meshtastic Device\nWiFi + MQTT enabled"]
        end

        subgraph wild["Public Mesh Node"]
            W_DEV["Stranger's device\nwith MQTT uplink"]
        end
    end

    A_DEV <-->| "LoRa radio\nhop_limit=3" | B_DEV
    A_DEV <-->| "LoRa" | GW_DEV
    A_DEV <-->| "LoRa" | W_DEV
    B_DEV <-->| "LoRa" | GW_DEV

    A_DEV <-->| "BLE" | A_PH
    B_DEV <-->| "BLE" | B_PH

    B_PH -->| "PATH 1: Phone GPS\nHTTP POST source=app" | API
    B_PH -->| "PATH 2: BLE Relay\nHTTP POST source=mesh_relay\nforwards ALL heard packets" | API
```

```

GW_DEV -->|"MQTT publish"| PUB
W_DEV  -->|"MQTT publish"| PUB
GW_DEV -.->|"MQTT publish\nif private configured"| PRIV

subgraph brokers["MQTT Brokers"]
  PUB["mqtt.meshtastic.org\nPublic broker"]
  PRIV["Private broker\nyour-broker.example.com"]
end

subgraph aervyx["Aervyx Backend"]
  API["REST API\n/api/track/position"]
  MSUB["MQTT Subscriber"]
  API --> RES["Resolve pilot_id\nauth token or\nmesh_device_id lookup"]
  MSUB --> RES
  RES --> TASK["Resolve task_id\nfrom active task"]
  TASK --> DB[("TrackingPoint")]
  DB --> SSE["SSE fan-out"]
end

PUB -->|"PATH 3a: subscribe\nmsh/+/+/position"| MSUB
PRIV -.->|"PATH 3b: subscribe"| MSUB

SSE --> MAP["Live Tracking Map"]

```

Path 1 -- Phone GPS (cellular)

The pilot's phone has cell service. The Aervyx mobile app sends the phone's own GPS coordinates directly to the backend API.

Step	What happens
GPS fix	Android/iOS location services provide lat/lon/alt
HTTP POST	POST /api/track/position With source=app
Auth	Bearer token identifies the pilot
Store + broadcast	Position saved, SSE pushed to live map

When it works: Pilot has cell service. Lowest latency, simplest path.

Path 2 -- BLE Relay (phone bridges mesh to API)

The phone is paired to a Meshtastic device via Bluetooth Low Energy. The device receives LoRa packets from the entire mesh network. The phone app reads those packets, parses the protobuf, and forwards every position to the backend.

Step	What happens
LoRa broadcast	Any mesh device broadcasts its position

Relay / rebroadcast	Other mesh nodes repeat the packet (up to <code>hop_limit</code>)
BLE delivery	Phone reads <code>fromRadio</code> BLE characteristic
Protobuf parse	App extracts <code>MeshPacket</code> -> Position (lat, lon, alt, speed)
HTTP POST	<code>POST /api/track/position</code> with <code>source=mesh_relay, device_id=!XXXXXXXX</code>
pilot_id lookup	Backend matches <code>device_id</code> to <code>users.mesh_device_id</code>
task_id lookup	Backend finds pilot's active task from event registration
Store + broadcast	Position saved, SSE pushed to live map

Key detail: This path forwards **all** mesh positions heard by the connected device, not just its own. If Pilot B's phone has cell service and Pilot A is beyond cell range but within LoRa range, Pilot A's position reaches Aervyx through Pilot B's phone.

Path 3 -- MQTT (mesh to broker to Aervyx subscriber)

A Meshtastic device with internet access (WiFi or Ethernet) and MQTT enabled acts as a gateway. It publishes every mesh packet it receives to an MQTT broker. The Aervyx backend subscribes to that broker.

Step	What happens
LoRa broadcast	Device broadcasts position via radio
Mesh relay	Other nodes rebroadcast (up to <code>hop_limit</code>)
Gateway receives	Any MQTT-enabled node with internet hears the packet
MQTT publish	Gateway publishes to broker topic <code>msh/US/2/e/position/...</code>
Aervyx subscribes	MQTT subscriber receives the <code>ServiceEnvelope</code> message
Protobuf parse	<code>ServiceEnvelope</code> -> <code>MeshPacket</code> -> Position
pilot_id lookup	<code>from_node</code> -> <code>!XXXXXXXX</code> -> <code>users.mesh_device_id</code>
task_id lookup	Active task resolution from event registration
Store + broadcast	Position saved, SSE pushed to live map

Public vs Private MQTT

Public broker (mqtt.meshtastic.org)

Your devices --> mesh --> ANY node with MQTT uplink --> broker
(yours OR a stranger's)

- Broader coverage: random hikers, other pilots, and public gateways all relay your positions for free
- Zero infrastructure: Meshtastic runs the broker
- Your positions are visible to anyone subscribing to the public broker

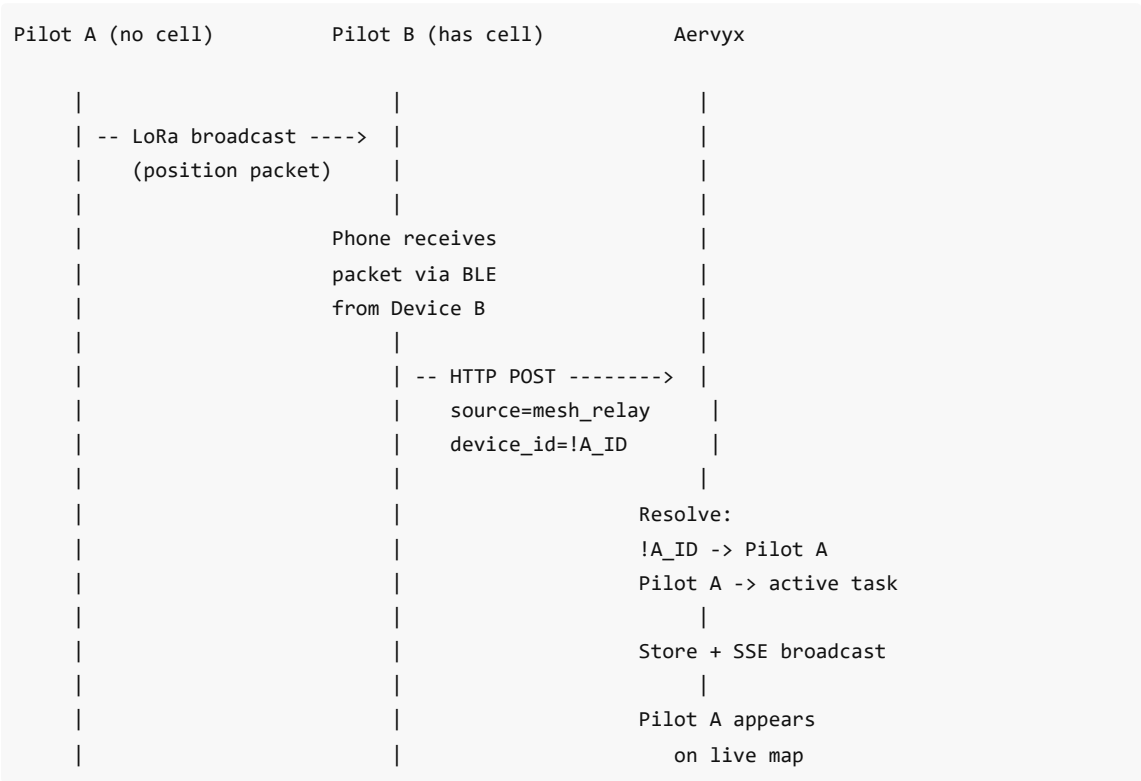
- You share bandwidth with all Meshtastic traffic worldwide

Private broker (self-hosted)

Your devices --> mesh --> only YOUR configured nodes --> your broker

- Private: only you see the data
- Dedicated bandwidth, no shared noise
- No benefit from public nodes -- strangers cannot relay to your broker because they do not have your credentials
- You must run and maintain the broker infrastructure

Scenario: Pilot A is out of cell range



Pilot A never had internet, but their position reached the map through the mesh network and Pilot B's phone acting as a relay.

Device ID auto-registration

When a pilot connects to a Meshtastic device via BLE in the mobile app, the app automatically registers that device's node ID (!XXXXXXX) against the pilot's account. No manual configuration is needed.

If the pilot switches to a different device, the new device ID overwrites the old one. This means the MQTT subscriber and BLE relay can always resolve incoming mesh positions to the correct pilot.

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sequenceDiagram
    participant Phone as Mobile App
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participant Device as Meshtastic Device

participant API as Aervyx API

participant DB as Database

Phone->>Device: BLE connect

Device-->>Phone: Config dump (includes myNodeNum)

Phone->>Phone: Format node ID: !XXXXXXX

Phone->>API: PUT /api/auth/mesh-device
{"mesh_device_id": "!abcd1234"}

API->>DB: UPDATE users SET mesh_device_id = '!abcd1234'
WHERE id = current_user.id

API-->>Phone: 200 OK

Note over Phone,DB: Future mesh positions from !abcd1234
now resolve to this pilot

Requirements for each path

Requirement	Path 1 (GPS)	Path 2 (BLE)	Path 3 (MQTT)
Phone has cell service	Required	Required	Not needed
Meshtastic device paired via BLE	Not needed	Required	Not needed
MQTT gateway node exists	No	No	Required
Device auto-registered to user	No	Yes (auto)	Yes (auto)
Pilot registered for active task	Required	Required	Required
Internet source	Phone	Phone	Gateway node